Bayesian Optimization of a Bioprocess across Reactor Scales

Modern bioprocesses underpin the production of pharmaceuticals, specialty chemicals, and biofuels. Their development is, however, slow and expensive: candidate operating conditions must be screened across multiple reactor scales, from small wells used for cheap parallel exploration up to a pilot reactor that mirrors the production environment. Each scale provides a different trade-off between experimental cost and how faithfully it predicts pilot-scale behavior. Choosing what to run next, and at which scale, is a sequential decision-making problem under uncertainty and limited budget. **In this project, students will probe a simulated multi-scale bioreactor and design a Bayesian Optimization (BO) algorithm that maximizes the amount of final product (Y) on the pilot scale while minimizing the total experimental cost across all scales.**

Task Description

The project is centered on a simulated bioprocess that can be operated at three reactor scales:

- **Microplate**: Cheapest per experiment but the noisiest and most biased.
- **Benchtop**: An intermediate trade-off between cost and information quality.
- **Pilot**: Most expensive and least noisy.

Each experiment is parameterized by a recipe $\mathbf{r} = [T, \text{pH}, F_1, F_2, F_3]^\top$ that contains the reactor temperature T, the reactor pH, and three feed rates F_1, F_2, F_3 for glucose, glutamine, and rich-media feeding (see Table 1 for admissible ranges). Each run returns an observation of the amount of final product Y (g/L) together with the experiment cost in EUR. The three scales share the same recipe space but differ in: (i) scale-dependent bias, (ii) observation noise on Y, and (iii) the cost per run (see Figure 1). The core

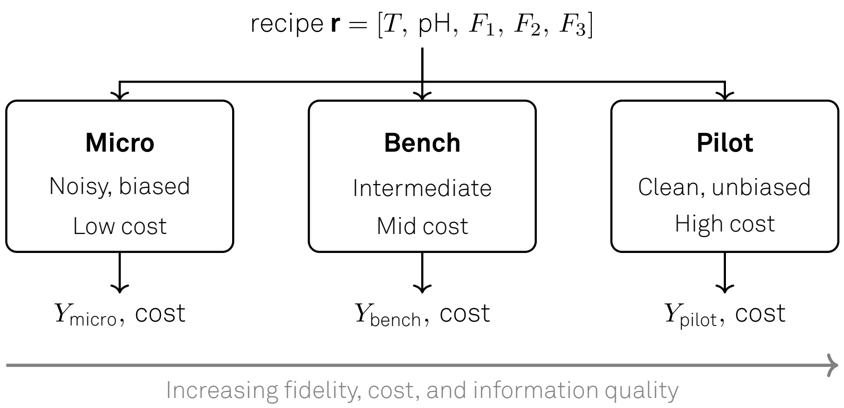

Figure 1: The three reactor scales available in the simulated bioprocess. All scales accept the same recipe and return a observation of the amount of final product Y and the experiment cost. Cheaper scales are biased and noisier estimates of pilot behavior

of the project is the design of a Bayesian Optimization loop that maximizes the pilot-scale performance (Y) while minimizing the total experimental cost across all scales.
BO maintains a probabilistic surrogate of the unknown objective, typically a Gaussian Process (GP) [2], and selects the next experiment by maximizing an ***acquisition function*** that trades off exploration of uncertain regions against exploitation of promising ones [1].

The System – Multi-Scale Bioreactor Simulator

The simulator runs on ***mlme26biosim.org*** and can be accessed via an internet browser UI or an API, with each group receiving its own login credentials. It accepts recipe inputs and returns product observations Y together with the corresponding experiment costs, while the exact process behind the simulation remains unknown. The recipe variables and their admissible ranges are listed in Table 1.

Table 1: Recipe variables and their admissible ranges. The same ranges are enforced server-side; recipes outside these bounds are clamped before simulation.

Variable	Range	Description
T	20 – 60 °C	reactor temperature
pH	3 – 9.5	pH
F_1	0 – 2 (g/L/h)	glucose feed rate
F_2	0 – 2 (g/L/h)	glutamine feed rate
F_3	0 – 2 (g/L/h)	rich-media feed rate

Mandatory tasks

The following tasks **have to be completed** in order to pass the project.

- Connect to the simulator API, run a small set of exploratory experiments, and visualize the dependence of Y on each recipe variable at each scale.
- Quantify the per-scale measurement noise and bias, and discuss how these properties influence the optimization strategy.
- Design and implement a Bayesian Optimization loop that selects, at each iteration, both the next recipe and the next scale.
- Run the BO loop multiple times and report the best pilot product amount obtained, the total cost spent, and the per-iteration trajectory of both quantities.
- Compare the BO loop against a non-adaptive baseline (e.g. Sobol or Latin Hypercube sampling spent across the same total budget) and discuss the result.

Additional tasks

Below are **suggested** additional tasks to obtain good or excellent grades for the project. We want to emphasize that students are encouraged to come up with their own ideas for additional investigations and not all of the suggestions below must be included for an excellent grade.

- Compare different acquisition functions (e.g. Expected Improvement versus Upper Confidence Bound) on the same simulator.
- Compare different surrogate model classes (multi-scale GP, independent per-scale GPs, Random Forest, ...).
- Compare the multi-scale surrogate against a pilot-only BO loop at equal total cost: does the cheap scale information actually help?
- Visualize the algorithm's per-iteration scale-selection pattern and discuss what drives the transitions between scales.
- Perform a global sensitivity analysis on the fitted surrogate to interpret which recipe variables drive product amount and how they interact.

Deliverables

The following materials have to be submitted **before the deadline** communicated via Moodle:

- **Final presentation** The presentation must be **7-10 minutes** followed by 20 min discussion to test understanding of concepts and code
- **Written report** to present and discuss the obtained results. You must use the supplied template on Moodle and write no more than **3-4 pages** (for the entire group). Highlight which group member worked on which section.
- **Source code** of your project. Please ensure that the code is executable and optionally add a short explanation of the structure (readme).
- **AI usage disclosure** for your submitted files. If generative AI tools were used in your submission, please list: (a) the AI model(s) employed, (b) which portions of code/report were AI-assisted, (c) the nature of the assistance, and (d) two detailed examples showing your iterative process of using AI to develop and refine code for this project.
- **Beat-The-Felix competition**: To participate, your submission must include a directory named 'Beat-The-Felix' containing a Python script called main.py. This script must:
  - Authenticate against the simulator API using credentials passed via environment variables or a config file.
  - Run a full BO campaign within the group's remaining budget.
  - Output, for each seeded run, the best pilot product amount (Y_{pilot}) found, the total cost spent, and the number of pilot evaluations performed.

  Points will be awarded if your algorithm achieves a higher mean best-pilot-product-amount than Felix's baseline at equal or lower mean cost, averaged over multiple seeded runs. Benchmarks are provided below.

Please ensure that all formal conditions (e.g. page limits, highlight responsible author) are satisfied, as we will deduct points for significant violations. Please submit all deliverables via Moodle.

Benchmarks

Table 2: Reference performance of Felix's baseline BO algorithm on the multi-scale bioreactor simulator, averaged over 10 independent seeded runs.

Metric	**Value**
Mean best pilot product amount (g/L)	14
Mean total cost (EUR)	15000

Responsible tutor

Please address questions to:

Name	**Contact**
Felix Brabender	felix.brabender@tu-dortmund.de

References

[1] Peter I. Frazier. ***A Tutorial on Bayesian Optimization***. July 8, 2018. arXiv: 1807.02811 [stat.ML]. URL: http://arxiv.org/abs/1807.02811 (visited on 05/18/2026). Pre-published.

[2] Carl Edward Rasmussen and Christopher K. I. Williams. ***Gaussian Processes for Machine Learning***. Adaptive Computation and Machine Learning. Cambridge, Mass: MIT Press, 2006. 248 pp. ISBN: 978-0-262-18253-9.